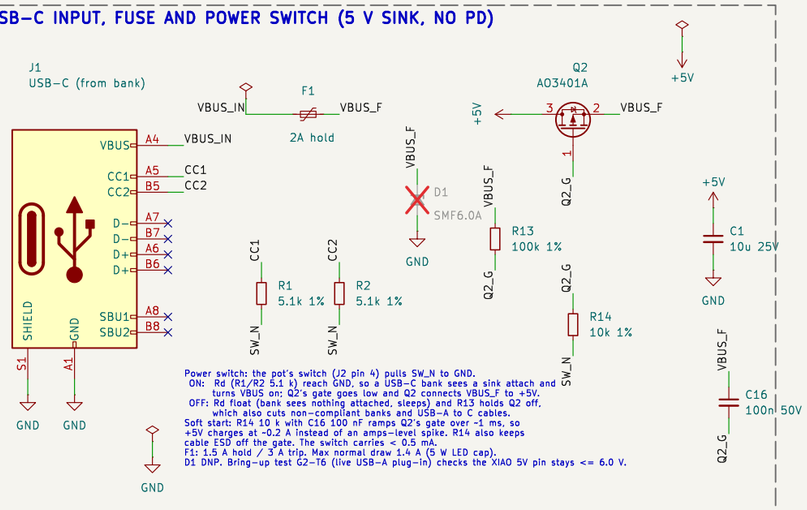
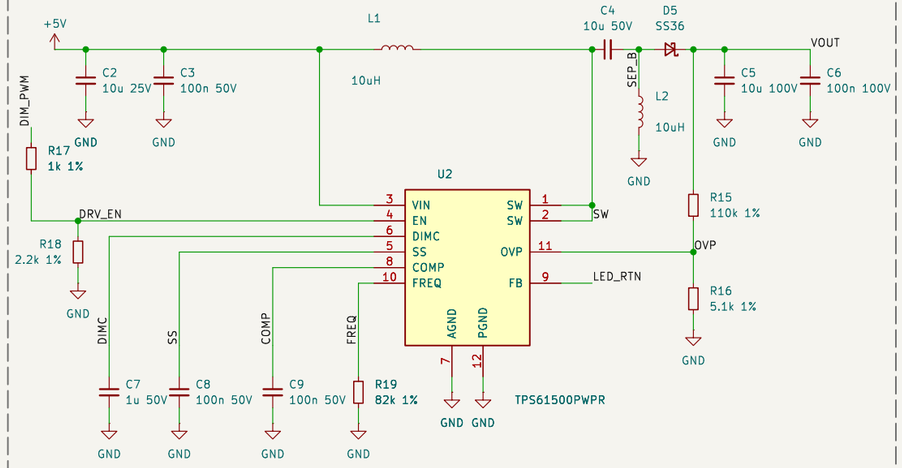
1 USB-C INPUT, FUSE AND POWER SWITCH (5 V SINK, NO PD)

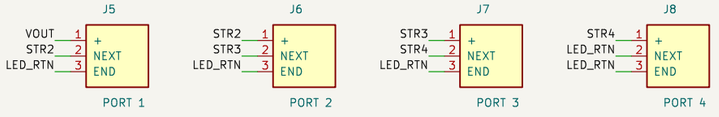


Power switch: the pot's switch (J2 pin 4) pulls SW_N to GND.
ON: Rd (R1/R2 5.1 k) reach GND, so a USB-C bank sees a sink attach and turns VBUS on; Q2's gate goes low and Q2 connects VBUS_F to +5V.
OFF: Rd float (bank sees nothing attached, sleeps) and R13 holds Q2 off, which also cuts non-compliant banks and USB-A to C cables.
Soft start: R14 10 k with C16 100 nF ramps Q2's gate over ~1 ms, so +5V charges at ~0.2 A instead of an amps-level spike. R14 also keeps cable ESD off the gate. The switch carries < 0.5 mA.
F1: 1.5 A hold / 3 A trip. Max normal draw 1.4 A (5 W LED cap).
D1 DNP. Bring-up test G2-T6 (live USB-A plug-in) checks the XIAO 5V pin stays <= 6.0 V.

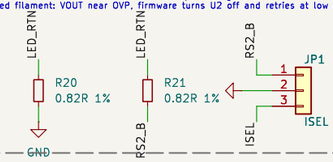
2 LED DRIVER: TPS61500 SEPIC (TI SLVA904), REGULATES 0.2 V ON FB



3 FILAMENT PORTS: ONE SERIES STRING, PUSH-IN TERMINALS



Fill ports in order from PORT 1. Each filament: + wire -> its port's +, - wire -> NEXT, except the LAST filament's - wire -> END (PORT 4's NEXT is END too). Longest string 24 V. WIRE PORTS ONLY WITH THE POWER SWITCH OFF: an open string holds VOUT near OVP. Open or reversed filament: VOUT near OVP, firmware turns U2 off and retries at low current.



I_max = 0.2 V / Rs. JP1 is a 2.54 mm header with one shunt:
2-3: R20 only, 244 mA; ISEL low.
1-2: R21 in parallel, 488 mA; ISEL high (R23).
none: 244 mA; ISEL high, so the firmware assumes 488 and under-drives (safe).

NOTES

Power: 5 V from the bank via Q2. LED power capped at 5 W in firmware (~1.4 A from the bank); 1.7 W lit at prototype brightness; idle ~4 mA with keep-alive, ~0.7 mA without (model: power_model.py).

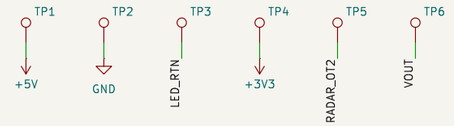
Pins (XIAO):
D1 GPIO3 ADC_POT ADC1_CH3
D2 GPIO4 VSENSE ADC1_CH4, VOUT / 13
D3 GPIO5 RADAR_OT2 wake, high = present
D4 GPIO6 UART_TX -> radar RX (115200 8N1)
D5 GPIO7 UART_RX <- radar OT1/TX
D6 GPIO21 ISEL_IO low = 244 mA (shunt 2-3), via R22
D7 GPIO20 DIM_PWM LEDC 19.5 kHz -> EN (analog dim)
D8 GPIO8 RADAR_MODE JP2 1-2 = manual, 2-3/none = radar
D10 GPIO10 KA_DRV keep-alive gate

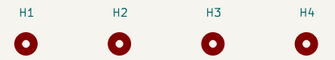
Assembly: JLCPCB fits everything incl. through-hole J2-J8 and the XIAO sockets; you plug in the XIAO and the radar. DNP: D1.

Bring-up (docs/g2_decisions.md): SEPIC at the extremes (one 3 V filament at 244 mA; 24 V string at the 5 W cap). filament temperature in the frame, bank wake via the switch.

SEPIC: +5V -> L1 -> SW (switch) ; C4 couples SW to SEP_B ; L2 SEP_B -> GND ; D5 SEP_B -> VOUT. Works for any string from one 3 V filament (~2.7 V) to 24 V, and C4 blocks DC: no current path from the bank to the LEDs when U2 is off.
f = 1.17 MHz (R19 82 k). OVP = 1.229 x (1 + 110k/5.1k) = 27.7 V (29.1 V worst case); switch sees VIN + VOUT + VD <= 35.1 V worst case (40 V abs max). L1 = L2 = 10 uH, 4.1 A sat. C5 is 100 V (X7S 1210) so it keeps its capacitance and > 2x margin at OVP.
Dimming: 19.5 kHz PWM on EN, filtered inside U2 by DIMC (C7) -> analog FB reference, I_LED = duty x I_max. EN low > 10 ms = shutdown (2 uA). R18 (2.2 k) holds EN below 0.4 V against GPIO20's ~45 k reset pull-up, so U2 stays off through boot and USB flashing.
COMP C9 100 nF. Lowest RHPZ 43 kHz (5 W into a 12 V string, 4.5 V in): keep crossover < 4.3 kHz. SLVA904 measured 1.7 kHz with 15 nF / 4.7 uH; 100 nF / 10 uH is lower still. Tune 15-470 nF on the bench. Checks: sepic_design.py (switch peak 2.2 A vs 3 A min).

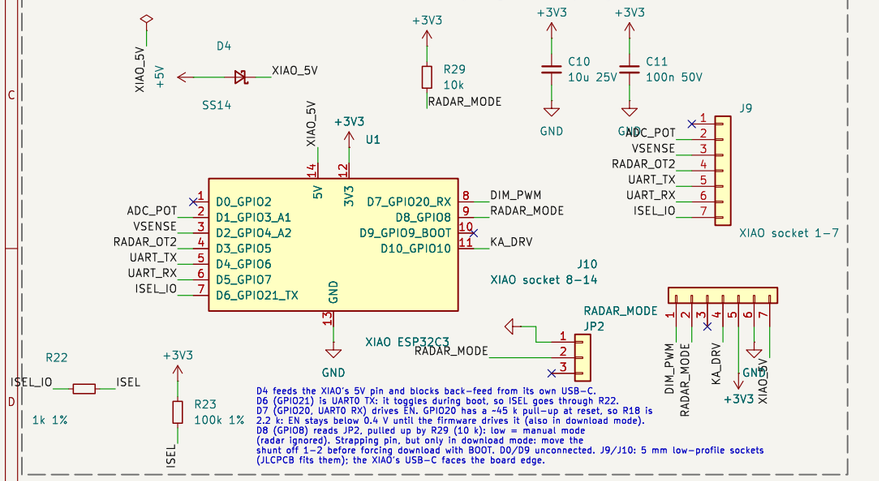
TP1 +5V TP2 GND TP3 LED_RTN (I = V / 0.82 R) TP4 +3V3 TP5 RADAR_OT2 TP6 VOUT





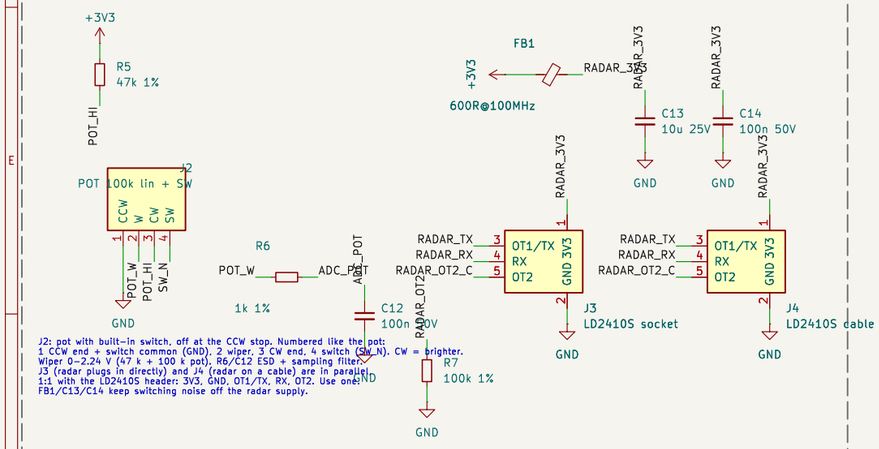
H1-H4: M3, place to suit the frame

4 MCU: SEEED XIAO ESP32C3 IN 5 mm SOCKETS (bought with pins)



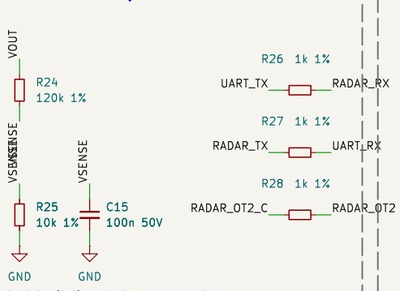
D4 feeds the XIAO's 5V pin and blocks back-feed from its own USB-C.
D6 (GPIO21) is UART0 TX: it toggles during boot, so ISEL goes through R22.
D7 (GPIO20, UART0 RX) drives EN. GPIO20 has a ~45 k pull-up at reset, so R18 is 2.2 k: EN stays below 0.4 V until the firmware drives it (also in download mode).
D8 (GPIO8) reads JP2, pulled up by R29 (10 k): low = manual mode (radar ignored). Strapping pin, but only in download mode: move the shunt off 1-2 before forcing download with BOOT. D0/D9 unconnected. J9/J10: 5 mm low-profile sockets (JLCPCB fits them); the XIAO's USB-C faces the board edge.

5 POT + POWER SWITCH, RADAR (SOCKET OR PIGTAIL)



J2: pot with built-in switch, off at the CCW stop. Numbered like the pot: 1 CCW end + switch common (GND), 2 wiper, 3 CW end, 4 switch (SW_N). CW = brighter. Wiper 0-2.24 V (47 k + 100 k pot). R6/C12 ESD + sampling filter.
J3 (radar plugs in directly) and J4 (radar on a cable) are in parallel, 1:1 with the LD2410S header: 3V3, GND, OT1/TX, RX, OT2. Use one.
FB1/C13/C14 keep switching noise off the radar supply.

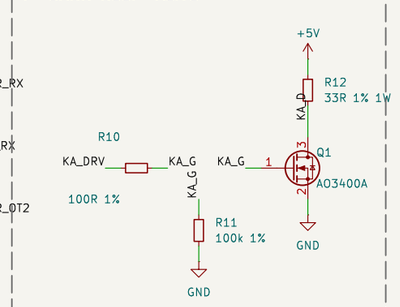
6 STRING VOLTAGE SENSE, RADAR LINE RESISTORS



VSENSE = VOUT / 13: 27.7 V (OVP) -> 2.13 V, inside the C3's ADC range at 11 dB. Firmware uses it for:
- P_LED = VOUT x I_set, capped at 5 W
- open string or reversed filament (VOUT > 25.5 V, near OVP): EN off, retry
- shorted output (VOUT ~ 0 while lit)
- each start: 10 % current first, VOUT checked against the learned window
- what is connected (one 12 V ~ 11 V, 3 x 3 V ~ 8.4 V)

0 uA when the driver is off (C4 blocks DC).
R26-R28: 1 k series on the radar lines (ESD on the cable/socket).

7 BANK KEEP-ALIVE



152 mA for 200 ms every 10 s (3 mA avg) only while the board draws less than the bank's threshold. Firmware setting can turn it off; R12 DNP = hard off.

Decisions: docs/g2_decisions.md. Values: hardware/tools/sepic_design.py
XIAO ESP32C3 (socketed), LD2410S radar (socket or pigtail), pot + power switch, keep-alive
Series filament string, TPS61500 SEPIC (TI SLVA904), 244/488 mA, 5 V USB-C input (no PD)

Rob Lim

Sheet: /
File: fgl_g2.kicad_sch

Title: Fused Glass Lighting controller

Size: A3	Date: 2026-10-28	**Rev: G2**
KiCad E.D.A. 9.0.7		Id: 1/1